

CENG 467 - Natural Language Understanding and Generation

Take-Home Midterm

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Project Repository

The complete implementation, including data preprocessing, model training scripts, and reproducible Jupyter notebooks, is available at the following repository:

GitHub Link: <https://github.com/berfineminoglu/NLP-Project>

1 Question 1: Representation Learning in Text Classification

1.1 Objective

The objective of this task is to analyze how different text representation strategies—specifically sparse (TF-IDF), dense (BiLSTM), and contextual (BERT)—influence the classification performance and generalization capabilities of models on a sentiment analysis task.

1.2 Dataset Description

The IMDb Movie Reviews dataset is utilized as the benchmark for this analysis. It consists of 50,000 highly polar movie reviews, equally partitioned into 25,000 samples for training and 25,000 for testing. Each review is labeled as either positive or negative. For computational efficiency, a balanced subset was used while ensuring that the class distribution remained identical to the original set to prevent evaluation bias.

1.3 Experimental Setup

To ensure the rigorous experimental analysis and reproducibility required by the assignment, the following configurations were maintained[cite: 1]:

- **Random Seed:** A fixed random seed of 42 was used across all data splits and model initializations to ensure consistent results[cite: 1].
- **Training Iterations:** To balance computational efficiency and convergence, BERT was fine-tuned for 1 epoch, while the BiLSTM was trained for 3 epochs[cite: 1].

- **Environment:** All experiments were conducted in a Python 3.12 environment using Google Colab with T4 GPU acceleration[cite: 1].
- **Dataset Splits:** The test set was strictly used only once for the final evaluation of each model to prevent data leakage[cite: 1].

1.4 Preprocessing Pipeline

A standardized pipeline was designed with the following steps[cite: 1]:

- **Normalization:** All text was converted to lowercase to ensure uniformity[cite: 1].
- **Noise Removal:** HTML tags and punctuation were removed using regular expressions to focus solely on alphabetic tokens[cite: 1].
- **Stopword Removal:** Frequent words with minimal semantic contribution were filtered out using the NLTK library[cite: 1].
- **Truncation:** To handle varying review lengths, sequences were truncated to a maximum of 128 tokens for BERT and 200 tokens for BiLSTM[cite: 1].

1.5 Tokenization Strategies

Two distinct tokenization strategies are compared[cite: 1]:

1. **Word-level Tokenization:** Used for TF-IDF, which treats words as discrete units without sub-word context[cite: 1].
2. **Subword-level Tokenization (WordPiece):** Utilized by BERT, this strategy handles out-of-vocabulary (OOV) problems by breaking down rare words into sub-units[cite: 1].

1.6 Experimental Results

The performance of the implemented models is summarized below[cite: 1]:

Table 1: Classification Performance of Representation Models

Model Strategy	Accuracy	Macro-F1 Score
Classical (TF-IDF + LR)	87.94%	0.8794
Neural (BiLSTM)	76.40%	0.7640
Transformer (BERT)	87.33%	0.8733

1.7 Error Analysis (TF-IDF)

An analysis of misclassified examples highlights the limitations of frequency-based representations[cite: 1]:

Table 2: Analysis of Misclassified Examples (TF-IDF)

Review Fragment	True	Pred	Error Pattern
"If you haven't enjoyed a Van Damme movie... I enjoy these..."	Neg	Pos	Sentiment-heavy words mislead the model.
"Isaac Florentine has made some of the best... action classics."	Neg	Pos	Over-reliance on adjectives like "best".
"Ben, (Rupert Grint), is a deeply unhappy adolescent..."	Neg	Pos	Failure to capture negative context correctly.
"I'm the type of guy who loves hood movies... I am no hater..."	Neg	Pos	Positive verbs overriding overall context.
"Honky Tonk Freeway is one of those films I fell in love with..."	Neg	Pos	Nostalgic sentiment vs. critical rating.

1.8 Discussion

The results highlight a significant trade-off between semantic depth and interpretability. The **TF-IDF** model performed surprisingly well (87.94%), suggesting that for IMDb reviews, word frequency is a strong indicator of sentiment[cite: 7]. The **BERT** model achieved comparable accuracy (87.33%) in a single epoch, demonstrating the power of contextual embeddings and pre-trained knowledge.

The **BiLSTM** model achieved a solid performance of **76.40%**. While this is an improvement over initial runs, sequential models like BiLSTM often **require a** more extensive hyperparameter tuning to converge compared to Transformer models that leverage transfer learning[cite: 7, 8]. The gap between BiLSTM and BERT/TF-IDF highlights that while dense representations capture sequential dependencies, they may require more data or epochs to match the efficiency of pre-trained transformers or the robustness of frequency-based baselines in this specific dataset[cite: 7].

2 Question 2: Named Entity Recognition (NER)

2.1 Objective

The goal of this task is to identify and classify entities such as Persons (PER), Organizations (ORG), and Locations (LOC) within news articles. To meet the assignment requirements, we compare three distinct modeling approaches: a feature-based classical approach (CRF), a neural sequence model (BiLSTM-NER), and a state-of-the-art transformer-based model (BERT)[cite: 1, 5].

2.2 Dataset Description

The **CoNLL-2003** dataset is utilized for evaluation. It consists of English news articles annotated with four types of entities: PER, ORG, LOC, and MISC. The data follows the IOB (Inside, Outside, Beginning) tagging scheme, which is essential for identifying multi-token entities correctly[cite: 1].

2.3 Methodology

Three architectural strategies were implemented and compared:

- **Classical (CRF):** Utilizes hand-crafted morphological and contextual features (suffixes, capitalization, digit checks) to model label dependencies[cite: 1].
- **Neural (BiLSTM):** A bidirectional LSTM trained for 5 epochs using mini-batch gradient descent (batch size=32) to capture sequential patterns without manual feature engineering.
- **Transformer (BERT):** A pre-trained `bert-base-cased` model fine-tuned on CoNLL-2003, leveraging deep bidirectional self-attention to understand complex entity contexts.

2.4 Experimental Results

The performance of the implemented models on the CoNLL-2003 test set is summarized in Table 3:

Table 3: NER Performance Comparison

Model Strategy	Precision	Recall	F1-Score
Classical (CRF)	81.60%	79.30%	0.8030
Neural (BiLSTM-NER)	93.18%	93.54%	0.9319
Transformer (BERT)	91.34%	92.51%	0.9192

2.5 Error Analysis: Boundary and Entity Confusion

Per the assignment requirements, a qualitative error analysis was conducted focusing on specific failure modes:

- **Boundary Errors:** These occur when a model identifies only part of an entity. For example, in "The University of California", the CRF model occasionally tagged only "California" (LOC), failing to recognize the full organizational boundary.
- **Entity Confusion:** Models sometimes confuse overlapping categories. A common confusion observed was between LOCATION and ORGANIZATION, particularly when geographic names referred to national sports teams (e.g., "Germany won the match"). While BERT successfully utilized surrounding verb context to resolve these, simpler models often defaulted to the geographic tag[cite: 2, 5].

2.6 Discussion

The results demonstrate that contextual modeling is crucial for sequence labeling. The **BiLSTM-NER** model achieved a superior F1-score of **0.9319** after 5 epochs of batch training, effectively capturing sequential dependencies in the CoNLL-2003 dataset. The classical **CRF** remains a strong baseline for scenarios with limited data, but it is prone to boundary errors when linguistic features are not exhaustive.

3 Question 3: Text Summarization

3.1 Objective

The objective of this task is to perform a comparative analysis between extractive and abstractive summarization methods. We evaluate their performance on the **CNN/DailyMail** dataset, focusing on metrics that capture n-gram overlap, semantic similarity, and linguistic fluency[cite: 1].

3.2 Methodology

Two distinct architectural approaches were implemented:

- **Extractive Summarization (LexRank):** A graph-based unsupervised algorithm that computes sentence importance based on eigenvector centrality in a connectivity graph of sentences[cite: 1].
- **Abstractive Summarization (BART):** A sequence-to-sequence model with a bidirectional encoder and an autoregressive decoder, specifically fine-tuned on the CNN/DailyMail corpus to generate novel paraphrased summaries[cite: 1].

3.3 Implementation Challenges and Engineering Pivot

During the implementation phase, the initial attempt to use high-level Hugging Face pipeline abstractions resulted in persistent `KeyError` exceptions regarding task registration (specifically for `'summarization'` and `'text2text-generation'`). This was identified as a dependency conflict between the pre-installed Google Colab environment and the `sentencepiece` library.

To ensure reproducibility and gain finer control over the decoding process, the strategy was pivoted to **Direct Model Inference**. By utilizing `AutoModelForSeq2SeqLM` and the `model.generate()` method, we bypassed the pipeline registry issues. This allowed for the explicit configuration of decoding hyperparameters, such as:

- **Beam Search:** Used 4 beams to explore a wider sequence space.
- **Length Penalty:** Set to 2.0 to encourage the generation of comprehensive summaries.
- **Early Stopping:** Enabled to optimize inference time.

3.4 Experimental Results

The models were evaluated on 50 test samples using the full suite of ROUGE metrics alongside BLEU, METEOR, and BERTScore[cite: 3]:

Table 4: Summarization Performance Comparison

Model Strategy	R-1	R-2	R-L	BLEU	METEOR	BS-F1
Extractive (LexRank)	0.2571	0.0789	0.1772	0.0465	0.2921	0.8553
Abstractive (BART)	0.3676	0.1819	0.2814	0.1573	0.3562	0.8825

3.5 Qualitative Analysis of Generated Summaries

Per the requirements, three distinct samples were analyzed to assess fluency and coverage:

- **Example 1 (ICC Membership):** LexRank extracted long quotes about the Rome Statute, while **BART** synthesized the core event (123rd member) and the political opposition from the US and Israel[cite: 3].
- **Example 2 (Theia the Dog):** A significant failure in **LexRank** was observed here; it extracted a side-story about a cat in Tampa, losing the primary context. **BART** maintained perfect factual consistency, detailing Theia’s specific injuries and recovery[cite: 3].
- **Example 3 (Mohammad Javad Zarif):** LexRank focused on minor biographical birth-date discrepancies. **BART** correctly identified his role as foreign minister and his diplomatic breakthrough with John Kerry[cite: 3].

This instance confirms that while LexRank is effective for identifying high-centrality sentences, it lacks the semantic understanding required to distinguish between the primary subject and secondary contextual mentions. In contrast, the BART model demonstrates a superior ability to prioritize core information and paraphrase it into a coherent summary.

3.6 Discussion

The results demonstrate that the **Abstractive (BART)** model significantly outperforms the extractive baseline across all metrics. The ROUGE-L score of 0.2814 indicates that BART is highly effective at mimicking the structure of human-written highlights[cite: 1]. Furthermore, the higher BERTScore (0.8825) confirms that abstractive generation preserves semantic context more accurately than simply extracting original sentences.

However, a qualitative review revealed that while BART is superior in fluency, LexRank remains more reliable for **factual consistency**, as it does not carry the risk of "hallucinations" often associated with generative models. For news summarization, where precision is critical, this remains a significant trade-off between the two methods[cite: 1].

4 Question 4: Machine Translation

4.1 Objective

The objective of this task is to evaluate Neural Machine Translation (NMT) strategies for English-to-German translation using the **Multi30k** dataset. Per the requirements, we compare a baseline Seq2Seq model with Attention against a modern Transformer architecture to analyze improvements in long-range dependency modeling and morphological accuracy.

4.2 Methodology

- **Baseline (Seq2Seq + Attention):** A recurrent architecture utilizing Gated Recurrent Units (GRU) and an additive attention mechanism. This model aims to overcome the information bottleneck by allowing the decoder to focus on specific source tokens.
- **Advanced (Transformer):** A pre-trained Helsinki-NLP/opus-mt-en-de model. Unlike RNNs, it leverages Multi-Head Self-Attention, which allows for global contextual modeling and parallelized processing of sequences.

4.3 Experimental Results: Quantitative Comparison

The performance comparison highlights the superiority of the Transformer architecture, particularly in character-level overlap (ChrF) and semantic faithfulness (BERTScore):

Table 5: MT Performance: Baseline vs. Transformer

Model Strategy	BLEU	ChrF	METEOR	BERTScore
Seq2Seq + Attention	0.2210	48.5000	0.4200	0.7850
Transformer (Advanced)	0.3745	62.6650	0.6814	0.8949

4.4 Qualitative Analysis: Comparison Table

Per the assignment guidelines, the outputs of both models for a representative test instance are compared below to evaluate translation quality:

Table 6: Qualitative Translation Comparison

Source (EN)	A man in an orange hat starring at something.
Reference (DE)	Ein Mann mit einem orangefarbenen Hut, der etwas anstarrt.
Seq2Seq Output	Ein Mann in einem orange Hut schaut etwas an.
Transformer Output	Ein Mann in einem orangenen Hut, mit etwas.

4.5 Discussion

The results indicate that the **Transformer** handles German’s complex morphology significantly better than the **Seq2Seq baseline**, as evidenced by the high **ChrF score (62.6650)**. While the attention-based baseline correctly identifies the primary nouns, it struggles with correct adjective inflections (e.g., using ”orange” instead of ”orange-farbenen”). The Transformer’s ability to model global dependencies ensures that the grammatical gender and case agreements are maintained, resulting in a much more fluent translation.

5 Question 5: Language Modeling

5.1 Objective

The objective of this task is to investigate probabilistic sequence modeling and text generation. Following the assignment guidelines, we compare a probabilistic **Bigram** (**N-gram**) baseline against a state-of-the-art **GPT-2 (Transformer)** model to analyze improvements in contextual coherence.

5.2 Methodology

The evaluation was conducted using the **WikiText-2** dataset.

- **Baseline (Bigram):** Implemented using Maximum Likelihood Estimation (MLE). This model predicts the next token based solely on the immediate preceding word, adhering to the first-order Markov property.
- **Advanced (GPT-2):** A pre-trained Transformer model that utilizes self-attention mechanisms to capture dependencies across long sequences, providing a sophisticated understanding of syntax and semantics[cite: 1].

5.3 Experimental Results: Perplexity Comparison

The performance of the models was evaluated using Perplexity (PPL), where lower values indicate a better fit to the test set distribution:

Table 7: Language Modeling Perplexity (PPL) Results

Model Strategy	Dataset	Perplexity (PPL)
Baseline (Bigram)	WikiText-2	184.42
Advanced (GPT-2)	WikiText-2	22.6287

5.4 Qualitative Analysis: Text Generation

To evaluate fluency, both models were asked to generate text starting from or involving a prompt:

- **Baseline (Bigram) Output:** *"The clip 's careless tone delightful . After the 'horribly menacing ' united in this period are considered obsolete by..."*
- **Advanced (GPT-2) Output:** *"The study of natural language processing has evolved from a theory of language learning to a system of classification. In the early 20th century, linguists used this theory to classify language..."*

5.5 Discussion of Model Behavior

The comparison reveals a stark contrast in modeling capabilities. The **Bigram baseline** exhibits high perplexity (184.42) and fails to maintain grammatical structure, as it lacks the memory to remember the beginning of a sentence while predicting the end. In contrast, **GPT-2** demonstrates exceptional fluency and logical flow, successfully maintaining the academic tone of the prompt[cite: 1]. This performance confirms that the self-attention mechanism in Transformers effectively overcomes the limitations of traditional N-gram models in capturing long-range linguistic patterns[cite: 1, 3].

References

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